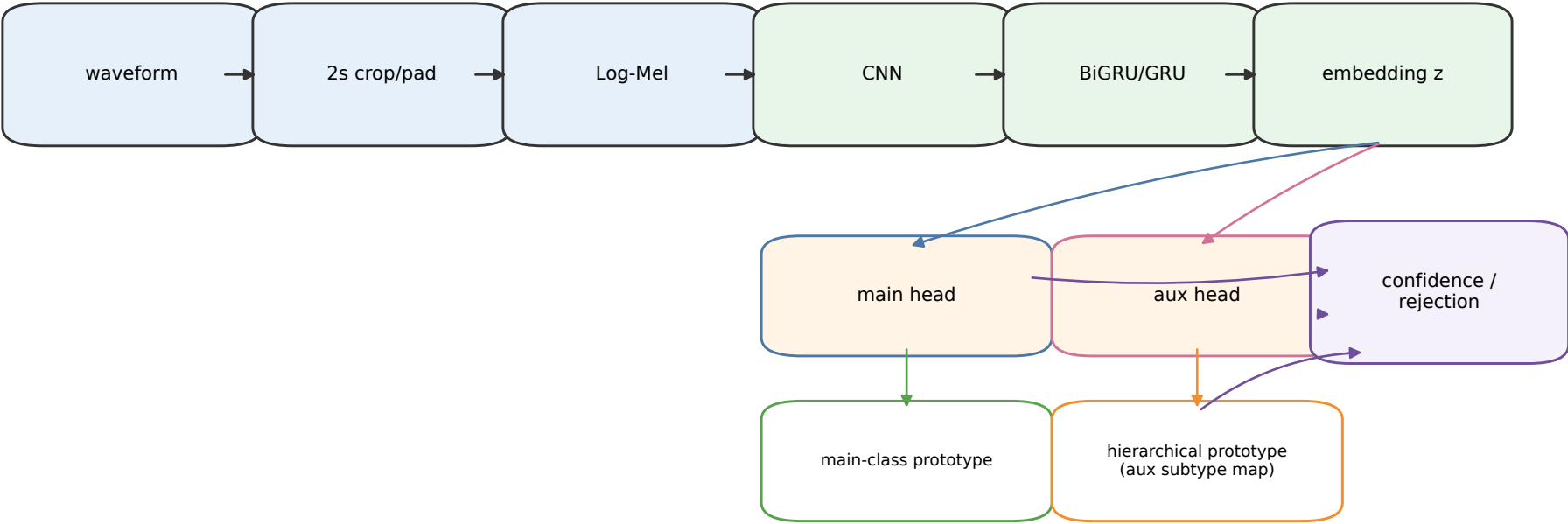


Overall method architecture

cap3x + 2-second Log-Mel CRNN with hierarchical subtype auxiliary supervision



Evaluation strata: clean | MODERATE_NOISE | EXTREME_STRESS | ALL_NOISY. Prototype/calibration artifacts are train/validation split specific.

Protocol source summary: cough; calm_grunt; feeding; stress_vocal; dry_cough; abdominal_cough; calm_grunt; feeding; frightened_stress; anxious_stress; cap3x expansion; 2-second Log-Mel; CRNN; 5 folds x 5 seeds; train-only prototypes; validation-only calibration; frozen test evaluation; no cross-fold or cross-seed mixing